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SHORT-PAPER

Make soil healthy again: Construction of ontology-compliant soil health knowledge graph with large language models

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Abstract

Soil health is fundamental to environmental sustainability and food security, yet relevant knowledge remains fragmented across diverse sources, hindering its effective application. Knowledge graphs (KGs) offer a robust solution by integrating disparate information into a structured, semantically rich format. Addressing this need, this paper presents an ontology-compliant soil health KG derived from domain literature, and the semi-automated, human-in-the-loop pipeline developed to construct it. Our pipeline leverages large language models to accelerate knowledge extraction, while incorporating expert oversight to ensure ontological compliance and accuracy. The resulting KG integrates unstructured knowledge into 11,715 RDF triples representing 2,018 entities, including 1,786 soil-related concepts. The KG's fidelity was confirmed by soil scientists through a validation process involving competency questions. The KG, supporting ontology, and the source code of the pipeline are made publicly available.

CCS Concepts

• **Computing methodologies** → **Semantic networks; Information extraction**; • **Information systems** → **Resource Description Framework (RDF)**.

Keywords

Knowledge graph, Soil health, Ontology-compliance, Large language models, Knowledge graph construction

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1 Introduction

Knowledge graphs (KGs) based on the Resource Description Framework (RDF) have emerged as a powerful means to store, integrate, and manage knowledge in a flexible yet semantically rich manner [8]. Already widely accepted and used across multiple domains—such as life sciences [22], cultural heritage [13], and e-commerce [20]—RDF KGs retain information in “triples” (subject,

predicate, object), linking diverse knowledge sources into a coherent graph that supports queries. Underpinning these KGs are ontologies [8], which provide pre-defined classes and properties for domain entities and their relationships. By offering standardized, formally defined terms, ontologies are crucial for ensuring consistent knowledge representation tailored to a specific domain.

Soil science represents one such domain where knowledge organization through semantic technologies, particularly KGs, can yield benefits. As in many other fields that have not yet widely used the Semantic Web as their primary paradigm for knowledge representation, knowledge about soil health remains fragmented, ambiguous and controversial across diverse scientific literature, policy documents, and practical guidelines, making it challenging to access, integrate, and apply effectively. While ontologies and KGs have been developed for specific aspects like the soil food web [10], soil properties and processes [3], and soil pollution management [7], an ontology-compliant KG dedicated to the multifaceted concept of soil health has not yet been realized.

Recent advances in Large Language Models (LLMs) [14] offer an opportunity to accelerate KG construction by automating knowledge extraction from unstructured text [27]. However, ensuring that LLM outputs align with established domain ontologies—a crucial step for creating robust and interoperable KGs—remains a challenge [20]. Therefore, a semi-automated, human-in-the-loop workflow [21] that combines the speed of LLMs with the precision of human expertise presents a more feasible and promising approach for developing specialized KGs.

Addressing this need, this paper presents two key contributions: (1) an ontology-compliant soil health KG (SHKG) derived from authoritative domain literature, and (2) the semi-automated, human-in-the-loop pipeline developed to construct it. Our pipeline leverages LLMs to accelerate knowledge extraction, while incorporating expert oversight to ensure ontological compliance and accuracy, demonstrating a reusable methodology for building expert-validated KGs in specialized scientific domains.

2 The ontology

The ontology for our KG emerged from a bottom-up construction process guided by the NeOn Methodology [25]. Our approach primarily involved reusing and merging existing ontological resources, which corresponds to Scenario 3 (Reusing ontological resources) and Scenario 5 (Reusing and merging ontological resources) of the NeOn. As shown in Figure 1, after LLMs generated candidate RDF triples, we first validated the completeness of the extracted entities and relationships. We then searched for standardized classes and properties from established ontologies to assign them.



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A variety of ontologies and schemas are in our scope. At a foundational level, we employ the Simple Knowledge Organization System (SKOS) Core [11] to structure the conceptual backbone of our KG. SKOS provides the framework for organizing soil-related terms (`skos:Concept`), their definitions (`skos:definition`), and the hierarchical relationships between them (`skos:broader` and `skos:narrower`). Complementing this, Dublin Core metadata terms [26] are utilized to reference and describe bibliographic resources (`dct:BibliographicResource`) and other entities (e.g., `dct:Policy` and `dct:Standard`) cited in the EU soil health report.

Beyond this basic structuring of concepts and metadata, expressing the rich and diverse relationships inherent in soil science requires a broader suite of ontologies. For these more advanced relational needs, we draw upon terms from several ontologies, including Agrontology [24], SemanticScience Integrated Ontology [4], Relation Ontology [23], SWEET Ontologies [19], etc. The complete ontology list can be found in the GitHub repository¹.

2.1 Ontology compliance

Although we strive to reuse established ontologies wherever possible, existing standards alone cannot fully express all relationships specific to our SHKG. In particular, we adopt Scenario 8 (Restructuring ontological resources) of the NeOn Methodology to define additional properties where none of the aforementioned ontologies adequately represents our domain needs. For instance, we define `she:sequesters`² (describing carbon sequestration processes in soils), `she:hasEquilibriumWith` (describing chemical balance), `she:hasDensity` and `she:hasBiomass` (describing the density and biomass of soil organisms).

By creating such an ontological extension that incorporates these new properties, we ensure that every entity and relationship in the KG is grounded in a formal definition, in accordance with the ontology-compliant approach [18]. In other words, all classes and properties in the ontology are utilized in our KG.

2.2 Relationship mapping from natural language to ontology properties

A key aspect of our ontology construction process is mapping natural language expressions of relationships to standardized ontology properties. Since human-authored texts often use varied phrasing for essentially the same conceptual relationship, normalizing these terms is crucial [2]. For example, when modeling the sentence,

Soil degradation can be defined as a decline in soil quality (Bone et al., 2010), resulting in the reduced functioning of the soil, we use `obo:RO_0002212` “negatively regulates” to represent the relationship between soil degradation and soil functioning. Regardless of whether the text uses terms like “reduce”, “decrease”, or “decline”, the fundamental relationship is captured by this ontology property. By consolidating these varied lexical forms into a finite set of ontology properties, we preserve semantic clarity and avoid the explosion of bespoke predicates.

¹<https://github.com/soilwise-he/soil-health-knowledge-graph>

²We prefix the newly defined entities and relationships in the KG with the `she` prefix, which represents SoilWise Horizon Europe.

Finally, after integrating all classes and properties to form the complete ontology for our KG, we validated its quality and consistency. We used the OOPS! (Ontology Pitfall Scanner!) web service [17] to automatically detect common modeling pitfalls. The analysis confirmed that our final ontology is free of detectable pitfalls.

3 Knowledge graph construction pipeline

Here we elaborate the steps of the pipeline for the construction of the SHKG, as illustrated in Figure 1.

3.1 Knowledge sources and conceptual model

We start with pre-processing the knowledge sources, which are the texts that will be inputted into the LLMs. The primary knowledge source for our KG is the report *Soil Monitoring in Europe* [6]. This report was selected because it consolidates the current scientific understanding of soil health and provides a visual conceptual model that defines the scope of soil health (see Figure 1.1 of the report). This model served as the primary guideline for constructing the upper layer of our SHKG. To address gaps in the primary report, such as limited coverage of soil threats like wind erosion and salinization, we used a supplementary report, *The state of soils in Europe* [5], as an additional knowledge source. In total, we selected 105 paragraphs from these two reports and defined one corresponding competency question (CQ) for each.

3.2 Knowledge graph generation using LLMs

In order to accelerate the construction of the SHKG, we leverage the LLM (in our case, GPT-4o [15]) to generate RDF representation for every selected paragraph from the knowledge sources. In particular, the process begins with the generation of candidate RDF triples from one paragraph of text through prompting. We invoke the model with only the text and KG generation prompt. After the LLM generates the first output, it will then check if any text has been stored as the literals of triples (after `skos:definition` or `skos:note`), but not yet extracting all potential entities and their relationships exhaustively. If so, we leverage a secondary extraction. LLM outputs at this stage are serialized in Turtle syntax, which are then parsed by `rdflib`³ to check for syntactical correctness. If errors are detected, a Turtle repair mechanism is initiated, feeding the triples with errors back to the LLM to generate a corrected response, which is then re-validated until there is no syntax errors.

3.3 Knowledge graph post-processing

Once syntactically correct, the KG segments undergo several post-processing steps. The first step is to review the generated triples to ensure that all entities and their relationships are correctly and completely extracted from the input text. After that, these candidate triples are manually aligned with ontologies, which we elaborated in Section 2. Then, we integrate these individual aligned KG segments with the RDF representation of the conceptual model. This results in a raw, unified graph that combines the high-level domain structure with detailed, extracted knowledge. This raw graph, however, typically contains duplicate entities and potentially conflicting relationships. Therefore, we then perform entity normalization and relation disambiguation to refine the KG.

³<https://github.com/RDFLib/rdflib>

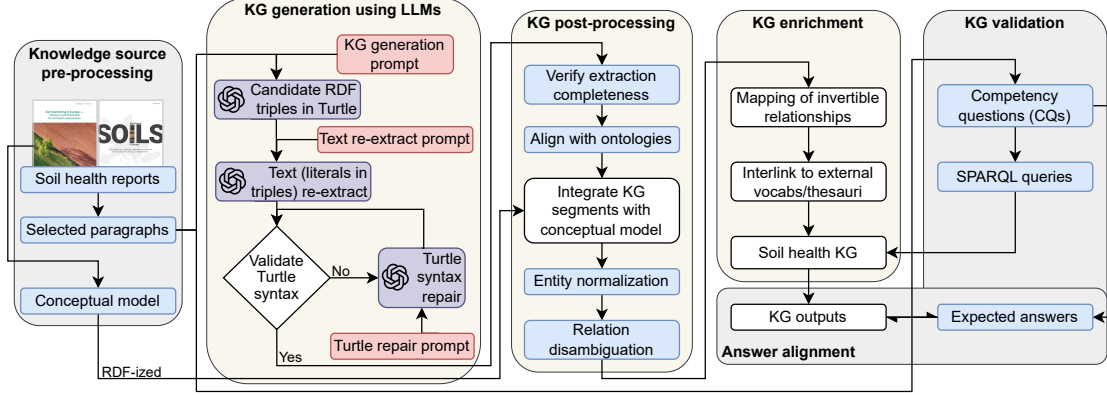


Figure 1: Knowledge graph construction pipeline. The blue modules indicate manual involvement, while other modules are automated. The purple modules represent the use of LLMs, which the outputs of LLMs are RDF triples serialized in Turtle. The red modules rely on manually designed prompts.

After integrating, each entity within the graph is required to have one preferred label (`skos:prefLabel`), and optionally one or more alternative labels (`skos:altLabel`). To merge duplicate concepts, we first generate vector embeddings for all entity labels using OpenAI’s text-embedding-3-large model [12]. We then calculate the cosine similarity between all entity pairs and cluster entities with scores above a set threshold. This process generates candidate groups for merging, where one preferred label is chosen, with all others as alternative labels.

After normalizing entities, we addressed conflicting relationships between entity pairs. First, we focus on SKOS hierarchical relationships. We identify two potential inconsistency patterns. The first one is transitive inconsistencies, which manifest as a clash between associative links and hierarchical links, such as the example in Listing 1. The second one is multiple broader concept issues, where a concept has more than one parent.

```

she:Soil skos:narrower she:SoilFunctions ;
          skos:related she:SOCContents .
she:SoilFunctions skos:narrower she:SOCContents .

```

Listing 1: Example of inconsistent hierarchical relationships.

Then, for other non-hierarchical relationships, we focus on identifying and resolving two potential forms of relationship inconsistency across URI-to-URI connections. First, we detect multiple relationship instances where different predicates connect the same subject-object pairs. Second, we identify bidirectional applications of the same predicate between concept pairs, excluding naturally symmetric predicates (e.g., `skos:related`).

For both normalization and disambiguation, the objective is not to automatically resolve all conflicts but to identify common patterns of inconsistency. The candidate clusters and detected conflicts are then presented to soil scientists for expert review and final resolution, ensuring the domain accuracy of the resulting KG.

3.4 Knowledge graph enrichment

Finally, we enrich the KG by materializing invertible relationships. If the predicate of a triple (subject-predicate-object) is invertible, we add the inverted triple (object-inversePredicate-subject) to the KG to ensure completeness. In addition, the KG is further enriched by interlinking to the controlled vocabularies and thesauri in the field of soil science, including AGROVOC [24], GloSIS [16], ISO

11074 [9], GEMET⁴ and INRAE Thesaurus [1], thereby aligning our KG with standard terminologies.

3.5 Human-in-the-loop validation

To validate the SHKG, we tested its ability to answer a set of 105 CQs that were defined a priori to guide the construction process. For each text paragraph fed into the pipeline, a corresponding CQ was formulated and reviewed by soil scientists. Each CQ was then translated into a SPARQL query to retrieve information from the KG. The query results were compiled into human-readable answers and compared against the source documents to verify accuracy and coverage. As the final step, soil scientists reviewed these answers for scientific validity, and the KG was adjusted based on their feedback to ensure it faithfully modeled the source knowledge.

3.6 Overview of the soil health KG

Figure 2 illustrates the high-level structure of the KG. The resulting SHKG comprises a total of 11,715 RDF triples, which describe 2,018 distinct entities. The majority of these entities (1,786) represent soil-related concepts, 493 of which are also matched in established soil-related thesauri. The remaining entities include 158 bibliographic references cited in the knowledge sources, along with other supportive entities. Structurally, the KG leverages 19 classes and 205 properties drawn from the selected ontologies, of which 45 properties are defined by us.

4 Conclusion and future work

In this paper, we addressed the need for structured soil health knowledge by presenting our two main contributions: an expert-validated, ontology-compliant SHKG and the semi-automated, human-in-the-loop pipeline used to create it. Our work provides a FAIR knowledge resource for the soil science community and showcases a reusable methodology that balances LLM-based extraction with the accuracy of human oversight, offering a practical approach for building KGs in specialized domains.

The KG is already serving as the backbone for a knowledge discovery portal⁵. In our future work, we will pay more attention to its potential contributions to the Semantic Web community. For

⁴www.eionet.europa.eu

⁵<https://voc.soilwise-he.containers.wur.nl/>

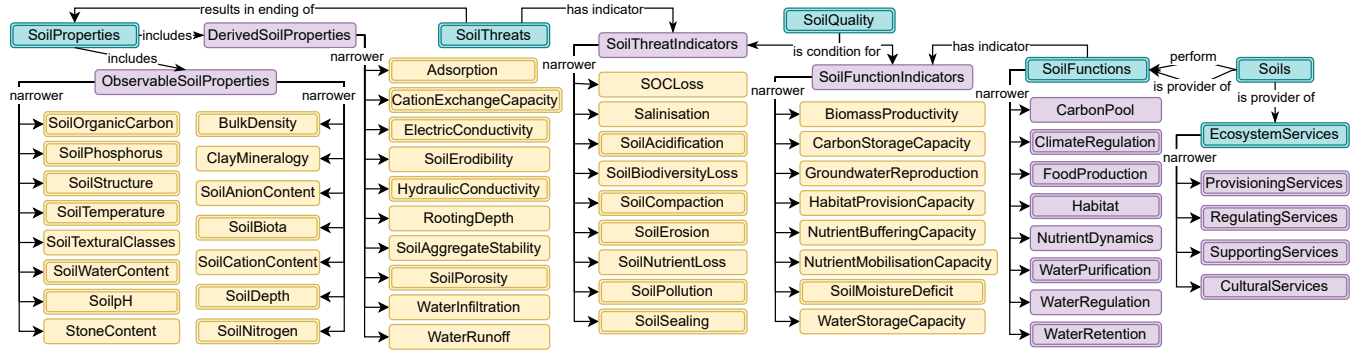


Figure 2: Overview of the high-level structure of the soil health KG, colors represent concepts at different echelons in the KG. The double-rounded rectangle indicates that the concept exists in other soil-related thesauri.

instance, the dataset of text segments paired with their ontology-compliant RDF triples can be released as a benchmark for text2KG research; considering the widespread use of reification with blank nodes to express conditions in scientific literature, it can become a challenging benchmark, especially for methods that leverage LLMs.

Resource availability statement: The KG, its supporting ontology schema and the validation dataset with CQs are available at <https://doi.org/10.5281/zenodo.14936019>. The AgroPortal instance is accessible at <https://agroportal.lirmm.fr/ontologies/SHKG>. The WIDOCO HTML page can be accessed at <https://soilwise-he.github.io/soil-health>, where all self-defined URIs are resolved to. The SPARQL endpoint is provided at <https://repository.soilwise-he.eu/sparql/>. Finally, the code and prompts for the KG construction pipeline can be found at <https://github.com/soilwise-he/soil-health-knowledge-graph>.

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